

# MTOjects / Toy Objects Optimisation

*Four-fold cross-validation on 40 low-foreground galaxies*

## 1. Aim

The MTOjects optimisation calibrates foreground-object masking against known injected Toy Objects while penalising unnecessary removal of galaxy pixels. It repeats the SEP experimental design on the same 40 validated galaxies, allowing the two algorithms to be compared under a common sampling framework.

## 2. Data and Toy Objects

The input list contains exactly 40 unique galaxies drawn from the 182-galaxy manifest. Each has complete bar/disc geometry and an accessible 3.6-micron FITS image. Six stars, compact clusters or elliptical galaxies are injected per image within the investigated region. Peaks span 5-25 robust-sigma and truth masks include model pixels above 8% of peak followed by one-pixel dilation.

| Toy property      | Rule                                                      |
|-------------------|-----------------------------------------------------------|
| Type mixture      | 50% star, 20% compact cluster, 30% galaxy                 |
| Star/cluster FWHM | 2-10 pixels                                               |
| Galaxy FWHM       | 5-22 pixels                                               |
| Galaxy axis ratio | 0.35-0.95                                                 |
| Placement         | Non-overlapping and wholly inside the investigated region |

## 3. Four-fold design

A fixed shuffle divides the same 40 names used by SEP into four disjoint folds of 10. In each rotation, MTOjects is optimised on 30 galaxies and evaluated on the excluded 10. Every galaxy therefore appears in a held-out set once. Each fold uses 40 Optuna trials: eight startup samples followed by 32 adaptive TPE trials.

| Phase                       | Images                         | Influences optimisation?              |
|-----------------------------|--------------------------------|---------------------------------------|
| Training                    | 30 per fold                    | Yes                                   |
| Held-out validation         | 10 per fold                    | No                                    |
| Common candidate evaluation | All 40, independent injections | Selects among four trained candidates |
| Final batch                 | All 182 manifest galaxies      | No                                    |

## 4. MTOjects processing model

MTOjects is run on the original science image for Toy Objects calibration. A baseline mask is first computed from the unmodified image. The same parameters are then applied to the injected image, and scoring uses only incremental pixels that were absent from the baseline mask. This separates Toy Object response from objects already detected in the galaxy.

incremental mask = injected-image mask AND NOT baseline mask

## 5. Objective

Recovery reward:  $R_{MT} = 0.45 \text{ mean F-score} + 0.35 \text{ mean toy recall} + 0.20 \text{ toy detection rate}$ .

Data-loss term:  $L_{MT} = 0.50 \text{ mean masked fraction} + 0.10 \min(\text{false-positive fraction}, 1)$ . Feasible score:  $S_{MT} = R_{MT} - L_{MT}$ ; Optuna minimises objective =  $-S_{MT}$ .

A toy is counted as detected when at least 50% of its truth pixels are recovered. A trial is recovery-infeasible when the incremental mask is empty, toy detection rate is below 25%, or mean toy recall is below 20%; it receives a large positive objective, with an empty mask receiving at least 100. Feasible trials retain the 15% worst-image masking cap. Final release requires common-40 toy detection of at least 50%, common-40 mean toy recall of at least 30%, and non-zero recovery in every held-out fold.

## 6. Optimised parameters and constraints

The search domains are the established constraints already implemented by the project MTOjects optimiser. Parameters not listed remain fixed at the processing-model defaults.

| Parameter       | Constraint                | Interpretation                                 |
|-----------------|---------------------------|------------------------------------------------|
| move_factor     | 0.0-1.0                   | Tree-node movement control                     |
| min_distance    | 0.0-1.0                   | Minimum hierarchy distance                     |
| gaussian_fwhm   | 0.0-5.0 pixels            | Gaussian smoothing applied before segmentation |
| bg_variance     | 0.0001-10000; step 0.0001 | Background variance supplied to MTOObjects     |
| minarea         | 1-80 pixels               | Minimum retained segment area                  |
| dilation_radius | 0-8 pixels                | Expansion of the accepted foreground mask      |
| max_area        | 20-3000 pixels            | Upper area gate protecting galaxy structure    |
| max_elongation  | 1.5-20                    | Shape gate rejecting extreme segments          |

## 7. Candidate evaluation and selection

Each fold contributes its best training parameter set. That candidate is scored on the fold's held-out 10 galaxies and on a common independent Toy Object realization across all 40 galaxies. The common set removes differences caused by comparing candidates on different held-out galaxies. The best feasible common-40 candidate is selected for the final batch only after the cross-fold recovery release gates are satisfied.

## 8. Outputs

| Artifact             | Review purpose                                                     |
|----------------------|--------------------------------------------------------------------|
| Fold trial summaries | Inspect every tested parameter combination and objective component |
| Held-out details     | Review per-galaxy generalisation and failures                      |
| Candidate comparison | Compare training, held-out and common all-40 metrics               |

| Artifact           | Review purpose                                        |
|--------------------|-------------------------------------------------------|
| Best JSON          | Preserve winning parameters and full provenance       |
| 182-galaxy summary | Confirm PNG status, runtime, mask fraction and errors |

## 9. Restart and monitoring

The workflow stores every Optuna study in SQLite and writes fold summaries incrementally. Supervisors detect unexpected launcher exits, reuse folds with all 40 trials, and restart incomplete folds or batches. Batch resume mode skips galaxies already present in the summary. Progress, remaining trials, rough ETA and expected completion are retained in terminal logs.

## 10. Interpretation

The preferred solution should combine stable parameters across folds with strong held-out recovery, controlled false-positive masking and no repeated dependence on extreme parameter bounds. The common 40-galaxy evaluation uses new injected objects but not a wholly external galaxy sample, so it measures injection-level generalisation within the selected low-foreground population.

## Systematic documentation review - 17 August 2026

**Review classification:** Updated - recovery objective, release gates and final winner

**Updates made:** The obsolete soft score was replaced by the implemented recovery reward/loss, infeasibility penalty and final release gates. Fold-3 selected parameters and common-40 metrics were recorded.

Current interpretation: MTOjects is no longer a zero-mask solution and is the recommended primary production method. It achieved higher four-fold held-out toy recall and detection than the corrected science-image SEP run, and its 182-galaxy masking tail was safer. Pixel precision remains a diagnostic rather than a catalogue-purity estimate because masks on real foreground sources outside injected toy truth are counted as false-positive area.

Authoritative combined outcome: SEP and MTOjects Masking Comparison and Recommendations 20260817.docx. Full optimisation history and final science-image SEP fold records: Foreground Masking Optimisation Results.xlsx.

Selected fold: 3. Common-40 score: 0.2325; mean toy recall: 50.7%; toy detection rate: 53.3%; mean masked fraction: 9.07%; maximum masked fraction: 14.63%; false-positive fraction: 9.05%; mean pixel precision: 0.31%.

Selected parameters: detect\_on=original; move\_factor=0.805164; min\_distance=0.215158;  
gaussian\_fwhm=0.429846; bg\_variance=0.000821033; minarea=5; dilation\_radius=3; max\_area=2987;  
max\_elongation=12.3811; exclude\_center\_pixels=8; alpha=1e-6.